

ASSESSMENT BRIEF

Programme	BSc Computer Science		
Module Title	Artificial Intelligence		
Module Code	CMP-N206-0		
Module Level	5	Assessment Type(s)	Practical coursework (Jupyter Notebook + Documentation)
Word Length / Duration	~1,500 words documentation (excluding references); Jupyter Notebook with code and visualisations	% contribution to module mark	50%
Deadline (date & time) for Submission	Thursday April 8, 2026	Format/Location of submission	Moodle (PDF documentation + .ipynb Notebook)
Assessment Feedback date:	Within 20 working days of submission		
Learning Outcomes This assessment has been designed to provide you with an opportunity to demonstrate your achievement of the learning outcomes listed below. By successfully completing this assessment, you will be able to: L01: Determine the characteristics of a given problem that an intelligent system must solve.		Employability and Professional Skills This assessment has been designed to provide you with an opportunity to demonstrate your achievement of the following skill(s) which is(are) critical in professional context and jobs. By successfully completing this assessment, you will be able to:	

L02: Evaluate whether a heuristic for a given problem is admissible/can guarantee optimal solution. L03: Make a probabilistic inference in a real-world problem using Bayes' theorem to determine the probability of a hypothesis given evidence. L04: Evaluate artificial intelligence solutions and methods in an ethical manner.	• Analyse complex problem spaces and identify suitable classical AI approaches. • Implement and document AI pipelines using industry-standard tools (Python, matplotlib). • Communicate results effectively through technical reporting, visualisation, and reflection. • Critically reflect on system design in relation to fairness, sustainability, and professional practice. • Demonstrate autonomy and problem-solving skills transferable to AI-based research.
Assessment Requirements Coursework 2 In this coursework you will design and implement a conversational AI agent that interacts with a user through text. Your agent must: • interpret user input • decide what to say • justify its responses • reason under uncertainty This coursework is about designing an intelligent agent that communicates rationally, not about training large language models. 2. The Scenario Your conversational agent helps users in a small domain. You may choose one domain (or propose your own): Examples: • Student course advisor • Travel assistant • Simple medical triage helper (non-diagnostic)	

- Library helper
- Game help assistant
- FAQ agent for a small system

The domain must be:

- narrow
- clearly defined
- safe and ethical

3. Core Requirements

A. Environment Model

Define:

- what information exists in the world
- what the user may ask
- what the agent knows

Example:

A travel agent knows:

- cities
- transport options
- prices

B. Percepts

User text input is the percept.

Your agent must interpret it using:

- rules
- keywords
- logic
- probabilities
- or a combination

C. Actions

Agent actions may include:

- answering a question
- asking for clarification
- refusing unsafe requests
- providing options

D. Knowledge Representation

Use at least one:

- rule-based logic
- fact database
- decision trees
- constraint rules
- simple ontology

Example rule:

IF user asks about exam dates

THEN retrieve academic calendar info

E. Inference

Use at least one:

- forward chaining
- backward chaining
- simple probabilistic reasoning
- Bayesian reasoning for intent detection

You must explain your reasoning process.

F. Handling Uncertainty

Your agent must demonstrate uncertainty handling.

Examples:

- confidence score
- asking clarification
- probabilistic intent choice

- saying “I am not sure”

G. Explainability (for stronger submissions)

Your agent must be able to answer:

“Why did you give that answer?”

Example:

“I answered this because your message matched the rule for timetable questions.”

H. Ethics & Safety

Discuss:

- bias
- misinformation risks
- user safety
- limitations

4. What You MUST NOT Do

- ✗ No training ML models
- ✗ No LLM fine-tuning
- ✗ No API reliance as core logic
- ✗ No black-box chatbots

AI tools may assist development, but your agent logic must be yours.

5. Deliverables

Notebook (.ipynb)

Include:

- environment model
- rules/logic/probabilities
- interaction examples
- comments explaining decisions

Report (~1500 words)

Include:

- domain description
- knowledge representation
- inference method
- uncertainty handling
- ethical reflection
- evaluation examples

6. Evaluation

Demonstrate at least:

- 5 example conversations
- cases where agent succeeds
- cases where it fails
- discussion of improvements

Formatting requirements:

- Documentation in 11pt Arial or Calibri, 1.5 line spacing, standard margins.
- Referencing in **IEEE style**.
- Files must be clearly labelled with student name and ID.
- No +10% leeway on word count.

Marking Criteria

Marks will be awarded using the University's categorical marking scheme. A short a video explaining the assessment criteria can be found at:

<https://www.youtube.com/watch?v=JrA0Dotq1p8>

This coursework rewards **understanding and effort**, not perfection. You will have the opportunity to provide evidence during an in-person demo towards the following grades:

42, 45, 48 (Basic Pass)

You show:

- Simple rule-based responses

- Limited reasoning
- Basic explanations
- Clear understanding of concepts

AI tools may have helped heavily, but you understand what your code does.

You meet the minimum learning outcomes.

52, 55, 58 (2:2 work)

You show:

- Several rules
- Some inference
- Some uncertainty handling
- Reasonable explanations
- *Some evidence of project management*

Not everything is fully correct or complete, but you clearly engaged and learned. Class participation counts.

62, 65, 68 (2:1 work, a Strong Submission)

You show:

- Clear knowledge base
- Correct inference method
- Proper uncertainty handling
- Strong explainability
- Good evaluation
- *Evidence of project management*

This is what a solid AI student submission looks like.

72, 75, 78 (First work Distinction)

You go **beyond** the specification, e.g.:

- Sophisticated reasoning
- Creative domain modelling
- Good probability use

- Strong ethical reflection
- Insightful limitations discussion
- *Evidence of reading beyond recommended material*
- *Excellent writing of a high academic standard*

You demonstrate deeper insight, creative and original elements and consideration of wider issues (real world application). You make sure ideas are supported by highly appropriate sources (not Generative AI).

82, 85, 92 (Upper First High Distinction)

Your work shows:

- Original and rigorous design
- Elegant reasoning structure
- Excellent evaluation
- Publishable-quality reflection
- Advanced but appropriate techniques

This is rare and not expected — it reflects exceptional attainment. Please have a chat with your Lecturer for guidance if you are attempting this classification. Your assignment might need to be adjusted and approved.

Subtle Inspiration for Strong Students (Not a Requirement)

Some advanced conversational agents:

- interact with other systems
- assist decision-making
- support autonomous agents

Students who explore such ideas thoughtfully may demonstrate deeper understanding.

Assessment Success Guidance

To achieve high marks, you should:

- *Frame your problem clearly as an **agent in an environment with objectives**.*
- *Choose evaluation metrics that connect directly to computational rationality.*
- *Engage with fairness, inclusivity, and sustainability, not just technical performance.*

- *Write with clarity, good structure, and correct IEEE referencing.*

Reading lists will be provided on Moodle.

This assessment links strongly with employability (critical problem analysis), sustainability (energy impact and fairness in AI), and internationalisation (applications in urban technology).

Assessment Guidance Support and Formative Feedback

Weekly support on class. Drop-in sessions and office hours for 1:1 feedback.

Contact for Queries/ who you can contact for further information or queries

*Module Leader: **Dr Arturo Araujo***

Contact details available via Moodle, along with office hours booking link

Ethical Requirements

Use only ethically approved datasets provided (if any).

Do not use identifiable or confidential data.

Reflect on fairness, inclusivity, and sustainability in your analysis.

Use of Artificial Intelligence (AI)

*The assessment is designed so that the use of AI during the assessment is **possible**. You must acknowledge any use of AI and appropriately cite all AI-generated outputs. Please make sure you read and understand the assessment guidelines and ask your Module Leader if you have any questions.*

Guidance on AI use:

- [*University AI Principles and Guidance for Senate*](#)
- [*Student Guidelines on the Use of AI*](#)

You may use AI tools for:

- debugging
- understanding concepts
- small code suggestions

But you must:

- understand your submission (every line of code and text)
- explain decisions
- document AI use

If you cannot explain your code, you cannot earn high marks.

Referencing

You must use **IEEE referencing** consistently throughout. Guidance can be found on the University Library Referencing Support page:
<https://library.roehampton.ac.uk/referencing>

Mitigating circumstances/late penalties

Sometimes circumstances outside of your control may affect your studies and might prevent you from submitting work on time or attending an exam.

The University offers the ability for students to request additional time to complete an assessment or to defer an examination to a later date. If you are finding yourself in such a situation, please speak to your Academic Guidance Tutor, the Roehampton Student Union (RSU) or someone in the [Wellbeing team](#) first, who can support you. Further details can be found on the [mitigating circumstances portal](#).

If you do not apply for or are not approved for Mitigating Circumstances, late penalties will apply. If work is submitted up to 14 days late, the mark will be capped at 50%; if it is over 14 days late, it will not be marked.

Resubmissions and Reassessment

If you are required to resubmit this assessment or take part in reassessment, you will be notified via Moodle and your student email. Please ensure you check both regularly. Any reassessment tasks will follow the same learning outcomes and criteria.

Submission Checklist

Before you submit, ask yourself:

Have I fully answered the assessment brief?

Have I met the word count and formatting requirements?

Is my referencing complete and accurate?

Have I declared any AI use honestly?

Have I proofread my work?

Am I submitting through the correct platform before the deadline?